Question	**Hypothesis**	**Sampling plan**	**Analysis Plan**	Rationale for deciding the sensitivity of the test for confirming or disconfirming the hypothesis	**Interpretation given different outcomes**	Theory that could be shown wrong by the outcomes
1. Is there evidence that the Sustainable Development Goals (2015) accelerated the adoption of low-carbon energy globally? **2.** Do the presence of national energy policies influence the share of low-carbon energy consumption?	**H0:** There is no statistically significant increase in the share of low-carbon energy consumption associated with the SDGs or policy strength. **H1:** The post-2015 SDG period is associated with a statistically significant increase in the share of low-carbon energy consumption globally. **H2:** Countries with stronger energy policies have higher shares of low-carbon energy consumption.	**1.** Dataset of countries annually over time (1992–2024), distinguishing between HICs, UMICs, LMICs. GDP per capita to ensure not just capturing richer countries already use more low-carbon energy. **2.** Data on government policies spending in low-carbon energy, and carbon pricing instruments will be gathered form the IEA (International Energy Agency).	**1.** Linear regression - SDG period (independent) vs low-carbon energy share (dependent)**.** Post-2015 = 1 Pre-2015 = 0 Baseline model: Low-carbon energy share = Post-2015 + GDP per capita + error term Compare pre-2015 vs post-2015 average levels (descriptive stats). Explore heterogeneity, differences across income groups. Robustness: test sensitivity to removing early years or changing pre/post sample periods. Check results remain consistent across specifications **2.** Linear regression model – number of government policies in energy spending or carbon taxation (independent) vs low carbon energy share (dependent), controlling for GDP per capita	We don't know how to do a power analysis so we're using a standardised $\alpha = 0.05$ - but emphasis placed on robustness across specifications rather than a single p-value - Effect sizes may be small due to gradual energy transitions - SDG influence may be indirect and lagged over time - Policy data may vary in measurement consistency across countries Sensitivity addressed by: comparing models with and without policy controls, testing robustness to sample restrictions (removing early years/incomplete policy coverage countries), evaluating whether results depend on inclusion of policy variable or GDP controls	**1.** If we see a statistically significant increase in energy share after the introduction of the SDGs we can reject the null that they haven't had a positive impact on low-carbon energy share which we can't confirm or deny otherwise **2.** If we see a statistically significant increase in energy share based on increases in government policies around energy spending and carbon taxation we can reject the null that these policies don't have an impact on low-carbon energy share which we can't confirm or deny otherwise	**Global governance effectiveness theory** (SDGs meaningfully influence national environmental outcomes) **Policy-driven energy transition theory** (formal national energy policies directly translate into measurable increases in low-carbon energy use) **Complementarity theory (SDGs + national policy interaction)** (if neither/only one effect observed, suggests weak interaction between global goals and domestic implementation